CCCA Meta-Atlas — Cell-State Manifold Analysis

Olbrecht et al., 2021 · High-Grade Serous Tubo-Ovarian Carcinoma · 10x · 3,936 malignant cells (6 samples / 5 patients)

GO / NO-GO GATES

Gate 1 GO

Mixing + continuity. 6 samples / 5 patients share ONE continuous manifold (single kNN component, no detached islands). Per-neighbourhood iLISI rises 1.00 -> 3.33 under Harmony (theta=10): sample-integrated, not a per-patient archipelago. Judged on the regenerated embedding, not a scalar alone.

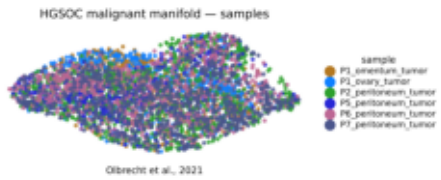
Gate 2 GO

Module geography (non-branching case). Gast/pSom co-vary (r=+0.44; +0.71 among non-cycling) and BOTH oppose Crest (pSom<->Crest r=-0.68); one developmental axis, Crest an inverse terminal gradient. Cell-cycle confound tested & rejected: anti-correlation survives excluding cycling cells; modules cc-free.

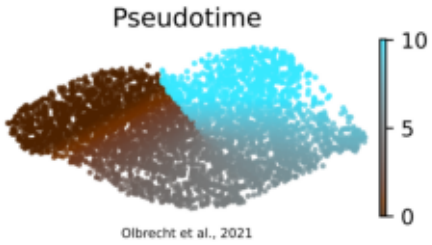
Gate 3 NA

Branching-only. No bifurcation (curve: 2 tips / 0 forks, fold-free) -> not tested. The non-branching case passes Gate 2 and reports Gate 3 as NA (distinct from a NO-GO where the pattern is absent).

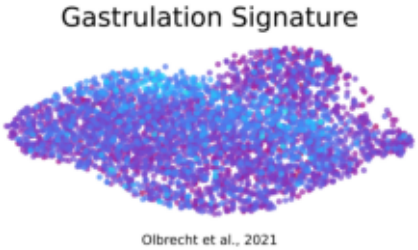
STEP FIGURES



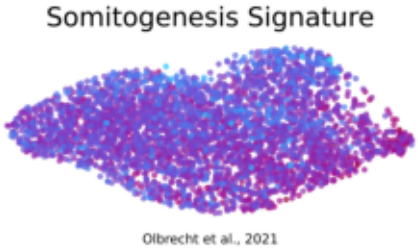
Samples — Harmony integration (Gate 1)



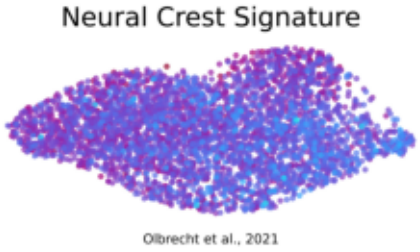
Pseudotime manifold (curve)



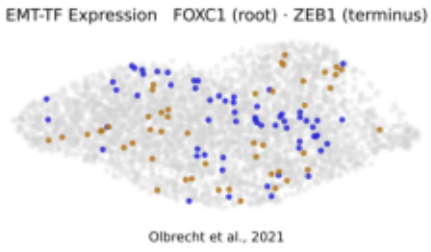
Gastrulation signature



Somitogenesis signature



Neural Crest signature



EMT-TF: FOXC1 (root) / ZEB1 (terminus)

PARAMETER CHOICES & RATIONALE

Choice	Value	Rationale
Sample threshold	>= 155 -> 6/10	Largest multiplicative count gap 384->63 (r6.10); cut at geometric mean, dropping the sparse tail (63/12/10/2, all normals included).
n_pcs	20	Scree elbow sits low (PC1-3 dominate); rejected as under-powering per eval audit, bounded to the Kurten(20)/Puram(40) precedent.
Harmony theta	10	Per-neighbourhood iLISI rises 1.00->1.17->3.20->3.33 monotonically; strongest integration, no collapse.
Island removal	0 cells	UMAP radius-graph (6x median NN) is a single component; the manifold is already continuous.
Geometry	Non-branching curve	Gast/pSom co-vary, both oppose Crest -> single axis. scFates curve, 2 tips / 0 forks, fold-free, stable across Nodes 15-30.
Root	tip 0 (Gast/pSom-high)	arg max(Gast+pSom-Crest) over tips; Crest is developmentally late. corr(x_lin, pseudotime)=+0.84.
Cell-cycle	Whole manifold kept	81%-cycling pocket at origin, but pSom<->Crest survives non-cycling (-0.63) and modules are cc-free -> developmental, not proliferative.
EMT-TF readout	FOXC1 -> ZEB1	Widened panel, 3% detection floor; FOXC1 root-relative (rho -0.06), ZEB1 terminus (rho +0.11). Matches the Kurten FOXC-at-root pattern.

VERDICT

Pattern REPRODUCES (non-branching). The embryonic programme recapitulates on a single continuous axis: a Gast/pSom-high origin resolving into an inverse, terminal Neural-Crest gradient (pSom<->Crest r=-0.68; along pseudotime Gast -0.33, pSom -0.06, Crest +0.40 terminal). Linear-pSom/Crest geometry (means Gast +0.029 / pSom +0.077 / Crest -0.020), not the Puram hub-and-spoke bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT: FOXC1 (root) -> ZEB1 (terminus).